

1.

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    printf("Hello Linux\n");
```

```
    return 0;
```

```
}
```

```
Last login: Fri Jul 31 14:27:39 on ttys000
[eeshayuktha@Eeshas-MacBook-Air ~ % ./hello
Hello, Linux
eeshayuktha@Eeshas-MacBook-Air ~ % █
```

2.

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    char name[30];
```

```
    printf("Enter your name: ");
```

```
    scanf("%s",name);
```

```
    printf("Welcome %s\n",name);
```

```
    return 0;
```

```
}
```

```
[eeshayuktha@Eeshas-MacBook-Air ~ % gcc printname.c -o name
[eeshayuktha@Eeshas-MacBook-Air ~ % ./name
Enter your name: Eesha
Welcome Eesha
eeshayuktha@Eeshas-MacBook-Air ~ % █
```

3.

```
#include<stdio.h>
#include<unistd.h>
```

```
int main()
{
    printf("Process ID = %d\n",getpid());

    return 0;
}
```

```
eeshayuktha@Eeshas-MacBook-Air ~ % nano process.c
eeshayuktha@Eeshas-MacBook-Air ~ % gcc process.c -o pro
eeshayuktha@Eeshas-MacBook-Air ~ % ./pro
Process ID = 49820
eeshayuktha@Eeshas-MacBook-Air ~ %
```

4.

```
#include<stdio.h>
#include<unistd.h>
```

```
int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

```
eeshayuktha@Eeshas-MacBook-Air ~ % nano fork.c
eeshayuktha@Eeshas-MacBook-Air ~ % gcc fork.c -o fork
eeshayuktha@Eeshas-MacBook-Air ~ % ./fork
Hello
Hello
eeshayuktha@Eeshas-MacBook-Air ~ %
```

5.

```
#include<stdio.h>
```

```
#include<unistd.h>
```

```
int main()
```

```
{
```

```
    pid_t pid;
```

```
    pid=fork();
```

```
    if(pid==0)
```

```
    {
```

```
        printf("I am Child Process\n");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("I am Parent Process\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
eeshayuktha@Eeshas-MacBook-Air ~ % nano pa.c
eeshayuktha@Eeshas-MacBook-Air ~ % gcc pa.c -o pa
eeshayuktha@Eeshas-MacBook-Air ~ % ./pa
I am Parent Process
I am Child Process
eeshayuktha@Eeshas-MacBook-Air ~ %
```

6.

```
#include<stdio.h>
```

```
#include<unistd.h>
```

```
int main()
```

```
{
```

```
    pid_t pid;
```

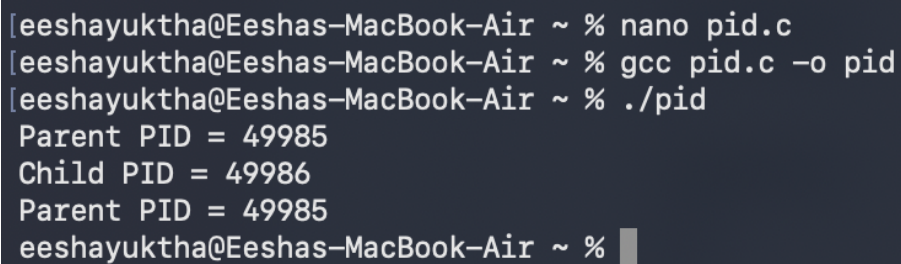
```

pid=fork();

if(pid==0)
{
    printf("Child PID = %d\n",getpid());
    printf("Parent PID = %d\n",getppid());
}
else
{
    printf("Parent PID = %d\n",getpid());
}

return 0;
}

```



```

eeshayuktha@Eeshas-MacBook-Air ~ % nano pid.c
eeshayuktha@Eeshas-MacBook-Air ~ % gcc pid.c -o pid
eeshayuktha@Eeshas-MacBook-Air ~ % ./pid
Parent PID = 49985
Child PID = 49986
Parent PID = 49985
eeshayuktha@Eeshas-MacBook-Air ~ %

```

7.

```

#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)

```

```

{
    printf("Child Running\n");
}
else
{
    wait(NULL);

    printf("Parent Running After Child\n");
}

```

```

[eeshayuktha@Eeshas-MacBook-Air ~ % nano prun.c
[eeshayuktha@Eeshas-MacBook-Air ~ % gcc prun.c -o prun
[eeshayuktha@Eeshas-MacBook-Air ~ % ./prun
Child Running
Parent Running After Child
eeshayuktha@Eeshas-MacBook-Air ~ % █

```

8.

```

#include<stdio.h>
#include<unistd.h>

```

```

int main()
{
    printf("Before exec()\n");

    execlp("ls","ls",NULL);

    printf("After exec()\n");

    return 0;
}

```

```

[eeshayuktha@Eeshas-MacBook-Air ~ % nano baexec.c
[eeshayuktha@Eeshas-MacBook-Air ~ % gcc baexec.c -o ba
[eeshayuktha@Eeshas-MacBook-Air ~ % ./ba
Before exec()
ba      Downloads      Movies      pa      printname.c      Public
baexec  fork               Music      pa.c    pro              readUsername
baexec.c fork.c            name      Pictures process.c         readUsername.c
Desktop hello          OS        pid     prun             SampleProgram.c
Documents Library      os1       pid.c   prun.c          un
eeshayuktha@Eeshas-MacBook-Air ~ % █

```

9.

```
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
```

```
int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        execlp("date","date",NULL);
    }
    else
    {
        wait(NULL);

        printf("Child Completed\n");
    }

    return 0;
}
```

```
eeshayuktha@Eeshas-MacBook-Air ~ % nano childcomp.c
eeshayuktha@Eeshas-MacBook-Air ~ % gcc childcomp.c -o cc
eeshayuktha@Eeshas-MacBook-Air ~ % ./cc
Tue Aug  4 20:58:24 IST 2026
Child Completed
eeshayuktha@Eeshas-MacBook-Air ~ %
```